

Handwritten exercises on GLMs

1 Exercise 1

Show that the Binomial distribution and the Gamma distribution belong to the exponential family of distributions, by identifying θ , A , ϕ , and the functions $b()$ and $c()$ (cf. p. 53 of the course slides). Using the properties of the distributions of the exponential family, give the expectation and the variance in each case.

Notation:

- If Y follows a binomial distribution $\mathcal{B}(m, p)$, then

$$P(Y = y; m, p) = \binom{m}{y} p^y (1-p)^{m-y}, \quad \text{for } y = 0, \dots, m.$$

- If Y follows a Gamma distribution $\Gamma(\mu, \nu)$, then

$$f(y; \mu, \nu) = \frac{\nu \exp(-\nu y / \mu) (\nu y / \mu)^{\nu-1}}{\mu \Gamma(\nu)}, \quad \text{for } y > 0.$$

2 Exercise 2

The beta distribution is used for situations where the measured variable takes values between 0 and 1, for example proportions.

We let Y follow a beta distribution with parameters μ and ϕ , where μ is the parameter of interest and ϕ is a nuisance parameter. The density function of Y is

$$f(y) = \frac{\Gamma(\phi)}{\Gamma(\phi\mu)\Gamma(\phi(1-\mu))} y^{\phi\mu-1} (1-y)^{\phi(1-\mu)-1}, \quad \text{for } 0 < y < 1.$$

Does this distribution belong to the exponential family? Justify your answer, and if yes deduce $E(Y)$ and $Var(Y)$.

3 Exercise 3

- Let $f(y; \theta)$ be the probability density function of a random variable Y , for a given parameter θ . Without using the exponential family structure, show that for $U(\theta; Y) = \frac{\partial}{\partial \theta} \log f(Y; \theta)$
 - a) $\mathbb{E}[U(\theta; Y)] = 0$,
 - b) $\mathbb{E}\left[\frac{\partial U(\theta; Y)}{\partial \theta}\right] = -\mathbb{E}[U^2(\theta; Y)]$.
- What condition do we have to impose on the support of Y to prove these two statements?

4 Exercise 4

In this exercise, we consider the problem of Maximum Likelihood estimation of the parameters of a Poisson random variable with a log link, i.e :

$$g(\mu_i) = \ln \{\mathbb{E}[Y_i \mid \mathbf{X}_i = \mathbf{x}_i]\} = \mathbf{x}_i^T \boldsymbol{\beta}$$

- a) Write the estimating (score) equations for the proposed model.
- b) Show that

$$\mathcal{J}(\boldsymbol{\beta}; \mathbf{y}) = \sum_{i=1}^n \exp\{\mathbf{x}_i^T \boldsymbol{\beta}\} \mathbf{x}_i \mathbf{x}_i^T.$$

- c) Write the formula for the Newton-Raphson algorithm.